A Bayesian Game of Online Accusations

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Abstract

Online accusations initiated by influential users often escalate into large-scale collective attacks despite substantial uncertainty about whether the target is guilty or innocent. This paper develops a Bayesian game capturing the strategic interaction between a Key Opinion Leader (KOL) who privately observes noisy information and a continuum of users who observe only the KOL's public accusation decision. In the baseline model, users decide whether to join an attack based on expected moral payoffs, social conformity, attention rewards, and the cognitive cost of resisting the crowd. We derive belief updating, characterize best-response thresholds for both the KOL and users, and provide conditions for pooling and separating Perfect Bayesian Equilibria (PBE). We then extend the model in a theory-consistent manner by allowing each user to receive a private signal about the target. This extension endogenizes partial participation: in equilibrium, the mass of attackers can be interior, and conformity incentives can generate multiple self-fulfilling "witch-hunt" equilibria. Additional extensions discuss heterogeneous users, multiple KOLs, and dynamic reputational penalties, linking the equilibrium logic to platform design and policy levers.

1 Introduction

Online platforms allow a single influential post to rapidly coordinate large numbers of users into a collective attack. Yet most users have no direct access to reliable evidence; they infer guilt from public signals such as an accusation made by a Key Opinion Leader (KOL). The central puzzle is that mass attacks can arise even when the target's guilt is highly uncertain. This paper provides a game-theoretic explanation: "digital witch-hunts" can be

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equilibrium outcomes of rational behavior under platform-generated attention rewards and social conformity incentives.

We formalize a Bayesian game in which a KOL privately observes noisy information about whether the target is guilty, decides whether to accuse, and thereby provides the only public signal observed by a continuum of users. Users then decide whether to attack or resist. The baseline model emphasizes two strategic complementarities: (i) users are more willing to attack when they expect others to attack; (ii) the KOL is more willing to accuse when crowd engagement amplifies attention and penalties for false accusations are weak. We then extend the model by introducing private user signals, which generates partial participation and clarifies how both information and coordination interact in sustaining large-scale attacks.

2 Baseline Environment

2.1 Players, Types, Prior

There are three components in the environment: a target whose type is chosen by Nature; a KOL, denoted L; and a continuum of identical users, represented by a typical user U.

The target's type is

$$T \in \{I, G\},$$

where I denotes innocent and G denotes guilty. The common prior is

$$\Pr(T = I) = p, \qquad \Pr(T = G) = 1 - p.$$

2.2 Timing and Information

The baseline game unfolds as follows.

1. Nature draws $T \in \{I, G\}$ according to $(p, 1-p)$.
2. The KOL privately observes a noisy signal $s_L \in \{s_L^+, s_L^-\}$.
3. The KOL chooses $a \in \{\text{Accuse}, \text{Not Accuse}\}$, publicly observed.
4. If $a = \text{Accuse}$, each user chooses $r \in \{\text{Attack}, \text{Resist}\}$.

The KOL signal is informative:

$$\Pr(s_L^+ \mid T = G) = \rho, \qquad \Pr(s_L^+ \mid T = I) = 1 - \rho, \qquad \rho > \tfrac{1}{2},$$

and hence

$$\Pr(s_L^- \mid T = G) = 1 - \rho, \qquad \Pr(s_L^- \mid T = I) = \rho.$$

Users receive no private signal in the baseline model; they condition only on the public action a.

3 Strategies and Beliefs (Baseline)

3.1 Strategies

A pure strategy for the KOL is a mapping

$$\sigma_L : \{s_L^+, s_L^-\} \to \{\text{Accuse}, \text{Not Accuse}\}.$$

A pure strategy for users is

$$\sigma_U : \{\text{Accuse}, \text{Not Accuse}\} \to \{\text{Attack}, \text{Resist}\}.$$

Only the action after Accuse is non-trivial.

Let $n_A \in [0, 1]$ denote the mass (fraction) of users who choose Attack after an accusation. Under symmetric pure strategies in the baseline (no user signals), $n_A \in \{0, 1\}$.

3.2 KOL Posterior Belief

After observing s_L, the KOL's posterior belief that the target is guilty is

$$\gamma(s_L) := \Pr(T = G \mid s_L).$$

Bayes' rule implies

$$\gamma(s_L^+) = \frac{\rho(1-p)}{\rho(1-p) + (1-\rho)p},$$
$$\gamma(s_L^-) = \frac{(1-\rho)(1-p)}{(1-\rho)(1-p) + \rho p},$$

and $\gamma(s_L^+) > \gamma(s_L^-)$ since $\rho > \frac{1}{2}$.

3.3 Users' Posterior Belief

Users observe only the KOL's action a. Define the posterior innocence belief after an accusation:

$$q := \Pr(T = I \mid a = \text{Accuse}).$$

By Bayes' rule,

$$q = \frac{\Pr(a = \text{Accuse} \mid T = I)\Pr(T = I)}{\Pr(a = \text{Accuse} \mid T = I)\Pr(T = I) + \Pr(a = \text{Accuse} \mid T = G)\Pr(T = G)}.$$

The likelihood terms are implied by σ_L:

$$\Pr(a = \text{Accuse} \mid T = t) = \sum_{s_L \in \{s_L^+, s_L^-\}} \mathbf{1}\{\sigma_L(s_L) = \text{Accuse}\}\Pr(s_L \mid T = t), \qquad t \in \{I, G\}.$$

4 Payoffs and Best Responses (Baseline)

4.1 User Payoff

After an accusation, a user chooses $r \in \{\text{Attack}, \text{Resist}\}$. Moral payoff is

$$M(T) = \begin{cases} M_G > 0, & T = G, \\ M_I < 0, & T = I. \end{cases}$$

Let n_A be the mass of attackers. Attacking yields social conformity payoff ϕn_A with $\phi > 0$ and attention gain $\alpha > 0$. Resisting yields social pressure cost $-\psi n_A$ with $\psi > 0$, and requires cognitive cost $K > 0$.

Given belief $q = \Pr(T = I \mid a = \text{Accuse})$,

$$EU(\text{Attack} \mid q, n_A) = qM_I + (1 - q)M_G + \phi n_A + \alpha,$$

$$EU(\text{Resist} \mid n_A) = -\psi n_A - K.$$

Define $\Delta_U(q, n_A) := EU(\text{Attack} \mid q, n_A) - EU(\text{Resist} \mid n_A)$:

$$\Delta_U(q, n_A) = qM_I + (1 - q)M_G + (\phi + \psi)n_A + \alpha + K.$$

Let $|M_I| = -M_I > 0$. Then

$$\Delta_U(q, n_A) = M_G - q(M_G + |M_I|) + (\phi + \psi)n_A + \alpha + K.$$

4.2 User Best Response Threshold

A user attacks iff $\Delta_U(q, n_A) \geq 0$, i.e.

$$q(M_G + |M_I|) \leq M_G + (\phi + \psi)n_A + \alpha + K.$$

Define

$$q^*(n_A) := \frac{M_G + (\phi + \psi)n_A + \alpha + K}{M_G + |M_I|}.$$

Then

$$\sigma_U(\text{Accuse}) = \begin{cases} \text{Attack}, & q \leq q^*(n_A), \\ \text{Resist}, & q > q^*(n_A). \end{cases}$$

4.3 KOL Payoff and Threshold

If the KOL accuses and a mass n_A attacks, his expected payoff is

$$EU_L(\text{Accuse} \mid s_L, n_A) = \gamma(s_L)(A_G + Bn_A) + (1 - \gamma(s_L))(A_I + Bn_A - \lambda D),$$

where $A_G > 0$ is attention payoff when the target is guilty, $A_I > 0$ is "black fame" attention even when innocent, $B > 0$ is amplification per unit crowd engagement, and λD is expected penalty when accusing an innocent ($\lambda > 0$, $D > 0$). If he does not accuse, normalize to

$$EU_L(\text{Not Accuse} \mid s_L) = 0.$$

Rewrite:

$$EU_L(\text{Accuse} \mid s_L, n_A) = \gamma(s_L)(A_G - A_I + \lambda D) + A_I + Bn_A - \lambda D.$$

Assume $A_G - A_I + \lambda D > 0$ and define

$$\gamma^*(n_A) := \frac{\lambda D - A_I - Bn_A}{A_G - A_I + \lambda D}.$$

Then

$$\sigma_L(s_L) = \begin{cases} \text{Accuse}, & \gamma(s_L) \geq \gamma^*(n_A), \\ \text{Not Accuse}, & \gamma(s_L) < \gamma^*(n_A). \end{cases}$$

5 Baseline Equilibria

We focus on PBE in pure strategies with symmetric user behavior. Under the baseline (no user signals), $n_A \in \{0, 1\}$, yielding "mob attack" or "universal restraint."

5.1 Pooling Accusation

If $\sigma_L(s_L^+) = \sigma_L(s_L^-) =$ Accuse, then the accusation carries no information and

$$q^{\text{pool}} = \Pr(T = I \mid a = \text{Accuse}) = p.$$

A pooling–mob PBE requires $p \leq q^*(1)$ and $\gamma(s_L^-) \geq \gamma^*(1)$ (and hence also $\gamma(s_L^+) \geq \gamma^*(1)$). A pooling–restraint PBE requires $p > q^*(0)$ together with $\gamma(s_L^-) \geq \gamma^*(0)$.

5.2 Separating Accusation

If $\sigma_L(s_L^+) =$ Accuse and $\sigma_L(s_L^-) =$ Not Accuse, then an accusation reveals $s_L = s_L^+$ and

$$q^{\text{sep}} = \Pr(T = I \mid s_L^+) = \frac{(1-\rho)p}{(1-\rho)p + \rho(1-p)}.$$

A separating–mob PBE requires $q^{\text{sep}} \leq q^*(1)$ and on-path KOL optimality $\gamma(s_L^+) \geq \gamma^*(1)$, with off-path beliefs chosen to deter deviation after s_L^-.

The baseline already shows how large B and small λD can make accusations frequent, while large (ϕ, ψ, α) and moderate K can make attacking individually optimal even under uncertainty. We now extend the model in a way that is both natural for online settings and analytically disciplined.

6 Extension I: Users Receive Private Signals (Partial Participation)

The baseline assumes users have no private information, so collective behavior is extreme ($n_A \in \{0, 1\}$) in pure strategies. A realistic extension is that each user receives a private, noisy signal about the target (e.g. personal experience, access to local facts, or exposure to alternative sources). This extension generates interior masses of attackers and clarifies how information interacts with coordination.

6.1 User Signals

Suppose that after the KOL accuses ($a = \text{Accuse}$), each user i privately observes a signal

$$s_i \in \{+, -\}.$$

The signal is informative about T:

$$\Pr(s_i = + \mid T = G) = \pi, \qquad \Pr(s_i = + \mid T = I) = 1 - \pi, \qquad \pi > \tfrac{1}{2},$$

and thus

$$\Pr(s_i = - \mid T = G) = 1 - \pi, \qquad \Pr(s_i = - \mid T = I) = \pi.$$

Signals are independent across users conditional on T.

The timing becomes: Nature draws T; KOL observes s_L and chooses a; if $a = \text{Accuse}$, each user observes s_i and chooses $r_i \in \{\text{Attack}, \text{Resist}\}$.

6.2 Public Belief and Private Posteriors

Let the public posterior innocence belief after an accusation remain

$$q := \Pr(T = I \mid a = \text{Accuse}),$$

as in the baseline (determined by the KOL strategy). Given q, a user updates further using s_i.

Posterior after $s_i = +$. Using Bayes' rule with prior q:

$$q_+(q) := \Pr(T = I \mid a = \text{Accuse}, s_i = +) = \frac{\Pr(s_i = + \mid I)\Pr(I \mid a = \text{Accuse})}{\Pr(s_i = + \mid I)\Pr(I \mid a = \text{Accuse}) + \Pr(s_i = + \mid G)\Pr(G \mid a = \text{Accuse})}.$$

Substituting $\Pr(s_i = + \mid I) = 1 - \pi$ and $\Pr(s_i = + \mid G) = \pi$:

$$q_+(q) = \frac{(1-\pi)q}{(1-\pi)q + \pi(1-q)}.$$

Posterior after $s_i = -$. Similarly,

$$q_-(q) := \Pr(T = I \mid a = \text{Accuse}, s_i = -) = \frac{\pi q}{\pi q + (1-\pi)(1-q)}.$$

Since $\pi > \frac{1}{2}$, we have $q_+(q) < q < q_-(q)$: a "+" signal makes the target look more guilty.

6.3 User Best Response with Private Information

Given signal $s \in \{+, -\}$, the user computes expected utility using $q_s(q)$ in place of q:

$$\Delta_U(q_s(q), n_A) = M_G - q_s(q)(M_G + |M_I|) + (\phi + \psi)n_A + \alpha + K.$$

Thus, for each $s \in \{+, -\}$, the user attacks iff

$$q_s(q) \le q^*(n_A), \qquad q^*(n_A) = \frac{M_G + (\phi + \psi)n_A + \alpha + K}{M_G + |M_I|}.$$

A natural class of symmetric equilibria in this extension is monotone in signals: users with $s_i = +$ (more suspicious of guilt) are at least as likely to attack as users with $s_i = -$. Because signals are binary, there are three canonical symmetric pure strategy profiles after an accusation:

$$\text{(Attack by both } + \text{ and } -), \quad \text{(Attack by } + \text{ only)}, \quad \text{(Attack by none)}.$$

6.4 Endogenous Mass of Attackers

If users follow a given signal-based rule, n_A is pinned down by the distribution of signals conditional on $a = \text{Accuse}$. Since $\Pr(T = I \mid a = \text{Accuse}) = q$ and $\Pr(T = G \mid a = \text{Accuse}) = 1 - q$,

$$\Pr(s_i = + \mid a = \text{Accuse}) = (1 - \pi)q + \pi(1 - q) =: \mu(q),$$

$$\Pr(s_i = - \mid a = \text{Accuse}) = 1 - \mu(q) = \pi q + (1 - \pi)(1 - q).$$

Therefore:

- If both signal types attack, then $n_A = 1$.
- If only + attacks, then $n_A = \mu(q) \in (0, 1)$.
- If nobody attacks, then $n_A = 0$.

6.5 Equilibrium Conditions (Given Public Belief q)

The next proposition gives sharp and correct conditions for the three canonical user outcomes after an accusation, taking the public belief q as given (which will be endogenized through the KOL strategy as in the baseline).

Proposition 1 (User Participation Regimes Given q)**.** *Fix a public posterior $q = \Pr(T = I \mid a = Accuse)$ and consider symmetric pure strategies where users condition on their private signal $s_i \in \{+, -\}$. Define $\mu(q) = (1-\pi)q + \pi(1-q)$ and posteriors $q_+(q), q_-(q)$ as above.*

1. *(**Universal attack**) A symmetric outcome in which all users attack ($n_A = 1$) is sequentially rational iff*
$$q_-(q) \leq q^*(1).$$
Since $q_-(q) \geq q_+(q)$, this condition ensures even the "−" users prefer to attack.

2. *(**Selective attack**) A symmetric outcome in which only users with $s_i = +$ attack and users with $s_i = -$ resist ($n_A = \mu(q)$) is sequentially rational iff*
$$q_+(q) \leq q^*(\mu(q)) \quad \textit{and} \quad q_-(q) > q^*(\mu(q)).$$

3. *(**Universal restraint**) A symmetric outcome in which all users resist ($n_A = 0$) is sequentially rational iff*
$$q_+(q) > q^*(0).$$
This condition ensures even the "+" users prefer to resist.

Proposition 1 shows that allowing private user signals yields an interior mass of attackers ($n_A = \mu(q)$) in a disciplined way: the crowd may be composed only of users who privately lean toward guilt.

6.6 Coordination and Multiple Equilibria

Because $q^*(n_A)$ increases in n_A (via conformity and social pressure), the extension admits the possibility of multiple self-fulfilling outcomes for the same public belief q: a larger anticipated n_A makes attacking more attractive, which validates a larger realized n_A. The binary-signal structure keeps this transparent: the "selective attack" regime can coexist with "universal attack" when parameters make the $-$ type close to indifferent.

Remark 1 (Why This Matters)**.** *The baseline can explain mass attacks purely from public coordination and incentives. The private-signal extension adds a new mechanism: even if many users privately doubt guilt, a sufficiently large conformity payoff can pull them into attacking, producing a high n_A that looks like "consensus" but is partly driven by strategic pressure rather than information.*

6.7 KOL Incentives with Private User Signals

The KOL's expected payoff from accusing depends on the equilibrium mass of attackers after an accusation. In the extended model, for each public belief q induced by σ_L, user equilibrium determines a mass

$$n_A \in \{0, \mu(q), 1\}.$$

Thus the KOL's threshold becomes

$$\gamma(s_L) \geq \gamma^*(n_A) = \frac{\lambda D - A_I - B n_A}{A_G - A_I + \lambda D},$$

but now n_A can be interior. This produces a sharp comparative statics: holding everything else fixed, the KOL is most willing to accuse when the user equilibrium mass is high (large n_A lowers $\gamma^*(n_A)$), and least willing to accuse when users are expected to restrain.

7 Extension II: Heterogeneous Users (Continuous Cutoffs)

A second natural extension is heterogeneity in users' moral or social sensitivity. This avoids the knife-edge "identical users" assumption and delivers interior participation even without private signals.

Let each user have a type θ that scales the conformity payoff (or moral cost). For concreteness, suppose the conformity term becomes $\theta\phi n_A$ and θ is distributed on $[\underline{\theta}, \overline{\theta}]$ with continuous CDF $F(\theta)$. The utility difference becomes

$$\Delta_U(q, n_A; \theta) = M_G - q(M_G + |M_I|) + (\theta\phi + \psi)n_A + \alpha + K.$$

For fixed (q, n_A), Δ_U is increasing in θ, implying a cutoff rule: there exists $\theta^*(q, n_A)$ such that users attack iff $\theta \geq \theta^*(q, n_A)$, where

$$\theta^*(q, n_A) = \frac{q(M_G + |M_I|) - M_G - \psi n_A - \alpha - K}{\phi n_A}.$$

The mass of attackers is then pinned down by a fixed point

$$n_A = 1 - F(\theta^*(q, n_A)).$$

This formulation generates interior n_A in a standard and correct way. Conformity still

creates strategic complementarity: higher n_A reduces the cutoff and increases participation, so multiple fixed points can arise for some parameter values, producing multiple equilibria in participation.

8 Extension III: Multiple KOLs (Competing Accusers)

This extension captures a basic empirical feature of real platforms: accusations do not come from a single influencer. Multiple KOLs may observe partial information, react strategically to each other, and compete for attention. When several KOLs accuse the same target, users interpret the accusations as corroboration and are more likely to join the attack. At the same time, KOLs may face strategic incentives to accuse earlier or more aggressively when the attention market rewards being a leader in a pile-on. We formalize these forces with a two-KOL Bayesian game and show how the baseline threshold structure generalizes to richer public histories.

8.1 Players, Signals, and Timing

There are two KOLs, indexed by $j \in \{1, 2\}$, and a continuum of users. Nature draws the target type $T \in \{I, G\}$ with prior $\Pr(T = I) = p$. Each KOL L_j privately observes a binary signal $s_j \in \{+, -\}$ with accuracy $\rho_j > \frac{1}{2}$:

$$\Pr(s_j = + \mid T = G) = \rho_j, \qquad \Pr(s_j = + \mid T = I) = 1 - \rho_j,$$

and signals are conditionally independent given T:

$$\Pr(s_1, s_2 \mid T) = \Pr(s_1 \mid T) \Pr(s_2 \mid T).$$

After observing s_j, KOL j chooses $a_j \in \{\text{Accuse}, \text{Not Accuse}\}$ simultaneously. Users observe the action profile $a = (a_1, a_2)$ and then choose whether to attack or resist.

8.2 Strategies and Beliefs

A (pure) strategy for KOL j is

$$\sigma_j : \{+, -\} \to \{\text{Accuse}, \text{Not Accuse}\}.$$

Users condition on the observed profile $a = (a_1, a_2)$; let

$$n_A(a) \in [0, 1]$$

denote the equilibrium mass of attackers after observing a.[1]

Users form a posterior about innocence given a:

$$q(a) := \Pr(T = I \mid a_1, a_2).$$

By Bayes' rule,

$$q(a) = \frac{\Pr(a \mid T = I)p}{\Pr(a \mid T = I)p + \Pr(a \mid T = G)(1 - p)},$$

where $\Pr(a \mid T = t)$ is induced by the KOL strategies and conditional independence:

$$\Pr(a \mid T = t) = \sum_{s_1, s_2 \in \{+,-\}} \mathbf{1}\{\sigma_1(s_1) = a_1\}\mathbf{1}\{\sigma_2(s_2) = a_2\} \Pr(s_1 \mid t) \Pr(s_2 \mid t).$$

Hence the public signal is richer than the baseline model: users can distinguish "no one accuses", "exactly one accuses", and "both accuse".

8.3 Corroboration and the Users' Attack Threshold

Users' payoff side remains the same as in the baseline model: given a belief $q(a)$ and an expected attacking mass $n_A(a)$, a representative user's net incentive to attack after observing a is

$$\Delta_U\big(q(a), n_A(a)\big) = M_G - q(a)\big(M_G + |M_I|\big) + (\phi + \psi)n_A(a) + \alpha + K.$$

Thus, users attack if and only if

$$q(a) \le q^*\big(n_A(a)\big) \quad \text{where} \quad q^*(n) := \frac{M_G + (\phi + \psi)n + \alpha + K}{M_G + |M_I|}.$$

A key comparative-static implication is immediate: for fixed strategies (σ_1, σ_2), if the profile a is more informative of guilt, then $q(a)$ is lower and attacking becomes more attractive.

To make the corroboration logic transparent, consider the natural separating strategies $\sigma_j(+) =$ Accuse and $\sigma_j(-) =$ Not Accuse for $j = 1, 2$. Then observing (Accuse, Accuse) reveals $(s_1, s_2) = (+, +)$, while observing (Accuse, Not Accuse) reveals $(+, -)$, etc. The

[1] Under baseline symmetric pure strategies, $n_A(a) \in \{0, 1\}$. If one combines this extension with the "private user signals" extension, then $n_A(a)$ can be interior. The present analysis does not require restricting $n_A(a)$ to $\{0, 1\}$.

posteriors become explicit:

$$q(\text{Accuse}, \text{Accuse}) = \Pr(T = I \mid +, +) = \frac{p(1-\rho_1)(1-\rho_2)}{p(1-\rho_1)(1-\rho_2) + (1-p)\rho_1\rho_2},$$

$$q(\text{Accuse}, \text{Not Accuse}) = \Pr(T = I \mid +, -) = \frac{p(1-\rho_1)\rho_2}{p(1-\rho_1)\rho_2 + (1-p)\rho_1(1-\rho_2)},$$

$$q(\text{Not Accuse}, \text{Accuse}) = \Pr(T = I \mid -, +) = \frac{p\rho_1(1-\rho_2)}{p\rho_1(1-\rho_2) + (1-p)(1-\rho_1)\rho_2},$$

$$q(\text{Not Accuse}, \text{Not Accuse}) = \Pr(T = I \mid -, -) = \frac{p\rho_1\rho_2}{p\rho_1\rho_2 + (1-p)(1-\rho_1)(1-\rho_2)}.$$

Since $\rho_j > \frac{1}{2}$, corroboration is powerful: $q(\text{Accuse}, \text{Accuse})$ is typically the smallest among these posteriors, so users are most willing to attack when both KOLs accuse. This delivers a micro-foundation for the common intuition that "two independent accusations" seem more credible than one.

8.4 KOL Payoffs with Attention Competition and Pile-On Complementarity

We now specify a minimal payoff structure that embeds strategic interaction between KOLs. KOL j receives attention benefits from accusing and from the ensuing crowd engagement, but those benefits may depend on whether the other KOL accuses as well. Let a_{-j} denote the other KOL's action. If KOL j accuses, his expected period payoff given his posterior $\gamma_j := \Pr(T = G \mid s_j)$ is

$$\begin{aligned} EU_{L_j}(\text{Accuse} \mid s_j, a_{-j}, n_A(a)) = \gamma_j \Big(A_G + B\, n_A(a) + \kappa\, \mathbf{1}\{a_{-j} = \text{Accuse}\} \Big) \\ + (1-\gamma_j) \Big(A_I + B\, n_A(a) + \kappa\, \mathbf{1}\{a_{-j} = \text{Accuse}\} - \lambda D \Big), \end{aligned}$$

and $EU_{L_j}(\text{Not Accuse} \mid s_j) = 0$. The new parameter κ captures how the other KOL's accusation affects j's attention payoff conditional on j accusing:

- $\kappa > 0$ (pile-on complementarity): joint accusations attract disproportionate traffic, so accusing becomes more profitable when the other accuses.
- $\kappa < 0$ (attention substitution): accusations crowd each other out, so accusing becomes less profitable when the other also accuses.

Rewriting, we obtain

$$EU_{L_j}(\text{Accuse} \mid s_j, a_{-j}, n_A(a)) = \gamma_j\big(A_G - A_I + \lambda D\big) + \Big(A_I + B\, n_A(a) + \kappa\, \mathbf{1}\{a_{-j} = \text{Accuse}\} - \lambda D\Big).$$

Assuming $A_G - A_I + \lambda D > 0$, KOL j accuses if and only if

$$\gamma_j \geq \gamma_j^*\big(n_A(a), a_{-j}\big) := \frac{\lambda D - A_I - B\, n_A(a) - \kappa\, \mathbf{1}\{a_{-j} = \text{Accuse}\}}{A_G - A_I + \lambda D}.$$

This generalizes the baseline threshold $\gamma^*(n_A)$.

8.5 Strategic Interaction Between KOLs

The expression for γ_j^* makes the interaction precise.

Complementarity (pile-on). If $\kappa > 0$, then

$$\gamma_j^*(n_A(a), \text{Accuse}) < \gamma_j^*(n_A(a), \text{Not Accuse}).$$

Thus, when j expects the other KOL to accuse, j requires weaker evidence to accuse. This generates a coordination force on the KOL side that mirrors the user-side coordination in the baseline model. In particular, the game may admit equilibria in which both KOLs accuse even when each observes only a moderately strong signal, because the anticipated pile-on makes accusation privately optimal.

Substitution (crowding out). If $\kappa < 0$, then joint accusation reduces marginal attention, so a KOL is less willing to accuse when the other accuses. In this case accusations can become strategic substitutes, and equilibria may feature only one KOL accusing (the one with stronger signal precision or lower expected penalties).

Corroboration feeds back into KOL incentives. Because a affects $q(a)$ and hence $n_A(a)$ through users' best responses, the KOLs' incentives also depend on whether users interpret the profile as corroboration. Typically, $n_A(\text{Accuse}, \text{Accuse}) \geq n_A(\text{Accuse}, \text{Not Accuse})$ because $q(\text{Accuse}, \text{Accuse}) \leq q(\text{Accuse}, \text{Not Accuse})$, which further lowers the threshold to accuse when the other accuses (via the $B\, n_A(a)$ term). Therefore, pile-ons can be amplified through two channels: (i) direct attention complementarity κ, and (ii) indirect crowd amplification $B\, n_A(a)$.

8.6 Equilibrium Patterns: Corroborated Witch-Hunts vs. Competitive Restraint

A Perfect Bayesian Equilibrium specifies $(\sigma_1, \sigma_2, \sigma_U)$ and beliefs $q(a)$ consistent with Bayes' rule on the path of play. While a full enumeration is lengthy, the model yields two clean qualitative regimes.

Corroborated witch-hunt equilibrium. Suppose (i) pile-on complementarity is strong ($\kappa > 0$), (ii) crowd amplification is strong (B large), and (iii) penalties are weak (λD small). Then the KOL thresholds γ_j^* are low after the other accuses, and users are most willing to attack when both accuse because $q(\text{Accuse}, \text{Accuse})$ is low. These forces can support an equilibrium in which both KOLs accuse on relatively weak evidence and users mount a mass attack after observing $(\text{Accuse}, \text{Accuse})$.

Competitive restraint / single-accuser outcomes. If attention is substitutive ($\kappa < 0$) or if joint accountability increases expected penalties when both accuse (e.g. an additional term $-\lambda_J D_J \mathbf{1}\{a_1 = \text{Accuse}, a_2 = \text{Accuse}\}$), then each KOL's incentive to accuse decreases when the other accuses. In this regime, equilibria may feature exactly one accuser (the one with higher precision ρ_j or higher moral/reputational concern), and users' posterior $q(\text{Accuse}, \text{Not Accuse})$ is higher than $q(\text{Accuse}, \text{Accuse})$, making attacks less likely.

8.7 Policy Interpretation

This extension highlights a policy-relevant tradeoff: corroboration improves information quality, but the attention market may distort incentives.

First, the corroboration channel is socially valuable: when accusations are independent signals, observing multiple accusers rationally lowers $q(a)$ and can help punish genuinely guilty targets.

Second, pile-on complementarity can induce excess accusation: even if each KOL individually has weak evidence, they may accuse because the other accuses, anticipating large engagement. This suggests that platforms should treat "multi-KOL pile-ons" as a special risk case: the same amplification that makes corroboration informative also makes strategic over-accusation profitable.

A natural platform lever is to reduce the incremental payoff from joint accusations by lowering κ and/or B when multiple high-reach accounts target the same individual in a short window (e.g. de-amplifying coordinated accusation cascades). Another lever is joint accountability: if two KOLs accuse and the target is later verified innocent, both face higher

penalties, which effectively raises γ_j^* in the joint-accusation region and makes pile-on equilibria harder to sustain.

9 Extension IV: Dynamic Reputational Penalties

This extension endogenizes accountability over time. The baseline model treats the expected penalty for falsely accusing an innocent target as an exogenous constant λD. In practice, reputational and platform penalties are often state-dependent: repeated false accusations increase the probability and severity of future punishment (e.g., demonetization, shadow-banning, credibility downgrades). We capture this mechanism with a simple two-period model in which (i) the KOL is forward-looking and (ii) the penalty for accusing an innocent target in period 2 is higher if the KOL was previously found to have made a false accusation.

9.1 Dynamic Environment and Timing

There are two periods $t \in \{1, 2\}$. Nature draws the target type $T \in \{I, G\}$ once at the beginning and the type remains fixed across periods. The common prior is $\Pr(T = I) = p$ and $\Pr(T = G) = 1 - p$.

At the beginning of each period t, the KOL observes a private signal $s_{L,t} \in \{s^+, s^-\}$ with the same accuracy $\rho > \frac{1}{2}$ as in the baseline model:

$$\Pr(s_{L,t} = s^+ \mid T = G) = \rho, \qquad \Pr(s_{L,t} = s^+ \mid T = I) = 1 - \rho,$$

independently across periods conditional on T. After observing $s_{L,t}$, the KOL chooses an action $a_t \in \{\text{Accuse}, \text{Not Accuse}\}$ which is publicly observed. After an accusation, users attack with mass $n_{A,t} \in [0, 1]$. For this extension, we treat $n_{A,t}$ as the (equilibrium) mass induced by the user side of the model given the public belief after $a_t = \text{Accuse}$. In particular, $n_{A,t}$ is allowed to be 0 or 1 (baseline users) or interior (e.g. under the private-signal extension), but the KOL anticipates it in period t when making his decision.

Between period 1 and period 2, verification may occur. If $a_1 = \text{Accuse}$, then with probability $\eta \in (0, 1]$ the true type becomes publicly revealed before period 2 (through investigation, official inspection, or credible fact-checking). If $a_1 = \text{Not Accuse}$, assume no investigation is triggered and hence no revelation occurs.[2]

[2]This assumption is without loss for the main discipline mechanism: what matters is that accusations increase the chance of eventual verification and reputational updating.

If verification occurs and $T = I$ while $a_1 = \text{Accuse}$, the KOL is recorded as having made a verified false accusation. This "strike" increases the penalty for accusing an innocent target in period 2.

9.2 State Variable and Reputational Penalty

Let the reputational state at the start of period 2 be $x \in \{0, 1\}$, where $x = 1$ indicates that the KOL has one verified false accusation on record. In period 1, the initial state is $x_1 = 0$.

The expected penalty from accusing an innocent target in period t is

$$\lambda(x_t)D,$$

where

$$\lambda(0) = \lambda, \qquad \lambda(1) = \lambda', \qquad \lambda' > \lambda > 0.$$

Thus, punishment is escalating: after a verified false accusation, the KOL faces a larger expected penalty if he accuses an innocent target again.

The state transition is simple. If the KOL accuses in period 1 and verification occurs and reveals innocence, the state becomes $x_2 = 1$; otherwise, $x_2 = 0$. Formally,

$$x_2 = \begin{cases} 1, & \text{if } a_1 = \text{Accuse},\ T = I,\ \text{and verification occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

Given $a_1 = \text{Accuse}$, verification occurs with probability η, so conditional on $T = I$ and $a_1 = \text{Accuse}$, we have $\Pr(x_2 = 1) = \eta$.

9.3 KOL Payoff in Each Period

Within period t, if the KOL accuses and the mass of attackers is $n_{A,t}$, his expected payoff conditional on his period-t posterior $\gamma_t := \Pr(T = G \mid s_{L,t})$ is

$$\begin{aligned} EU_{L,t}(\text{Accuse} \mid s_{L,t}, n_{A,t}, x_t) &= \gamma_t(A_G + Bn_{A,t}) + (1 - \gamma_t)\big(A_I + Bn_{A,t} - \lambda(x_t)D\big) \\ &= \gamma_t\big(A_G - A_I + \lambda(x_t)D\big) + A_I + Bn_{A,t} - \lambda(x_t)D, \end{aligned}$$

and $EU_{L,t}(\text{Not Accuse} \mid s_{L,t}) = 0$.

The KOL discounts period 2 by $\delta \in (0,1)$, so his total expected payoff is

$$EU_L = EU_{L,1} + \delta\, EU_{L,2}.$$

Crucially, the period-1 decision affects the distribution of x_2, which affects the effective penalty term $\lambda(x_2)D$ in period 2.

9.4 Period-2 Problem: A State-Dependent Threshold

Fix the state $x_2 \in \{0,1\}$. In period 2, the KOL's problem is identical to the baseline model except that the penalty parameter is $\lambda(x_2)$ rather than λ. Hence the period-2 accusation threshold is

$$\gamma_2(s_{L,2}) \geq \gamma_2^*(n_{A,2}; x_2) := \frac{\lambda(x_2)D - A_I - Bn_{A,2}}{A_G - A_I + \lambda(x_2)D}.$$

Because $\lambda(1) = \lambda' > \lambda(0) = \lambda$, we have

$$\gamma_2^*(n_{A,2}; 1) > \gamma_2^*(n_{A,2}; 0)$$

whenever $A_G - A_I + \lambda(x)D > 0$ for both states. Therefore, a "struck" KOL (with $x_2 = 1$) requires strictly stronger evidence (higher posterior of guilt) to accuse in period 2.

9.5 Period-1 Problem: Dynamic Incentive and Effective Threshold

In period 1, accusing yields the usual within-period payoff plus the discounted continuation value, which depends on whether the KOL is likely to enter the high-penalty state $x_2 = 1$.

Let $V(x)$ denote the KOL's expected ex ante continuation value at the start of period 2 when the state is $x \in \{0,1\}$, before observing $s_{L,2}$, under equilibrium behavior in period 2 (including the induced user mass $n_{A,2}$). Then the continuation value after choosing $a_1 =$ Accuse in period 1 is

$$\mathbb{E}[V(x_2) \mid a_1 = \text{Accuse}, s_{L,1}],$$

while after $a_1 =$ Not Accuse it is simply $V(0)$ because x_2 stays at 0 (no strike can be recorded).

We can write, using $\Pr(T = I \mid s_{L,1}) = 1 - \gamma_1$ and $\Pr(T = G \mid s_{L,1}) = \gamma_1$,

$$\begin{aligned}\mathbb{E}[V(x_2) \mid a_1 = \text{Accuse}, s_{L,1}] &= \gamma_1 V(0) + (1-\gamma_1)\Big(\eta V(1) + (1-\eta)V(0)\Big) \\ &= V(0) + (1-\gamma_1)\eta\big(V(1) - V(0)\big).\end{aligned}$$

Since higher penalties reduce future gains from accusing, it is natural (and holds under the

present structure) that

$$V(1) \leq V(0),$$

so $V(1) - V(0) \leq 0$. Therefore, accusing in period 1 imposes an additional expected dynamic cost proportional to the probability of being innocent and verified, $(1 - \gamma_1)\eta$.

The KOL accuses in period 1 iff the total (current + discounted continuation) payoff is non-negative:

$$EU_{L,1}(\text{Accuse} \mid s_{L,1}, n_{A,1}, x_1 = 0) + \delta\Big(\mathbb{E}[V(x_2) \mid a_1 = \text{Accuse}, s_{L,1}] - V(0)\Big) \geq 0.$$

Substituting the expression above,

$$EU_{L,1}(\text{Accuse} \mid s_{L,1}, n_{A,1}, 0) + \delta(1 - \gamma_1)\eta\big(V(1) - V(0)\big) \geq 0.$$

Because $V(1) - V(0) \leq 0$, the second term is weakly negative and reduces the incentive to accuse in period 1, especially when the KOL believes the target may be innocent (small γ_1), verification is likely (large η), and the future matters (large δ).

To express this as a threshold condition, define the dynamic adjustment term

$$\Omega := \delta\eta\big(V(0) - V(1)\big) \geq 0.$$

Then the period-1 accusation condition becomes

$$EU_{L,1}(\text{Accuse} \mid s_{L,1}, n_{A,1}, 0) \geq \Omega(1 - \gamma_1).$$

Using the baseline expression

$$EU_{L,1}(\text{Accuse} \mid s_{L,1}, n_{A,1}, 0) = \gamma_1(A_G - A_I + \lambda D) + A_I + Bn_{A,1} - \lambda D,$$

the inequality is

$$\gamma_1(A_G - A_I + \lambda D) + A_I + Bn_{A,1} - \lambda D \geq \Omega(1 - \gamma_1).$$

Rearranging terms yields the dynamic accusation threshold:

$$\gamma_1 \geq \widetilde{\gamma}^*(n_{A,1}) := \frac{\lambda D - A_I - Bn_{A,1} + \Omega}{A_G - A_I + \lambda D + \Omega}.$$

This expression is correct and mirrors the static threshold $\gamma^*(n_A)$, except that the dynamic term Ω enters both numerator and denominator and strictly increases the threshold whenever

$\Omega > 0$.

9.6 Main Conclusions

The extension yields clean, testable comparative statics.

Proposition 2 (Dynamic Discipline Raises the Standard of Evidence)**.** *Assume* $A_G - A_I + \lambda D > 0$, $\lambda' > \lambda$, *and* $V(0) \geq V(1)$ *so that* $\Omega \geq 0$. *Then the KOL's period-1 accusation threshold satisfies*

$$\widetilde{\gamma}^*(n_{A,1}) \geq \gamma^*(n_{A,1}),$$

with strict inequality when $\Omega > 0$. *Moreover,*

$$\frac{\partial \widetilde{\gamma}^*}{\partial \delta} > 0, \qquad \frac{\partial \widetilde{\gamma}^*}{\partial \eta} > 0, \qquad \frac{\partial \widetilde{\gamma}^*}{\partial (V(0) - V(1))} > 0,$$

and

$$\frac{\partial \widetilde{\gamma}^*}{\partial n_{A,1}} < 0.$$

The proposition says: (i) forward-looking KOLs require stronger evidence to accuse in period 1 compared with the static model; (ii) the evidence standard rises when the future matters more (δ), when verification is more likely (η), and when the reputational wedge is larger ($V(0) - V(1)$), i.e. when a "strike" severely harms future payoff opportunities; (iii) crowd amplification still works in the opposite direction: a larger expected mass of attackers $n_{A,1}$ lowers the threshold, partially offsetting dynamic discipline.

9.7 Interpretation and Link to Platform Policy

The object η captures the platform's verification intensity: how likely accusations are to be checked and resolved before the public moves on. Increasing η strengthens dynamic discipline by raising the expected chance that a false accusation becomes publicly verified and recorded. The gap $\lambda' - \lambda$ and the value loss $V(0) - V(1)$ capture escalating sanctions: demonetization, reduced reach, credibility labels, or permanent trust-score reductions. The discount factor δ captures how much the KOL values long-run reputation relative to short-run attention.

This extension therefore delivers a precise mechanism for why "credible enforcement over time" can shift the equilibrium away from witch-hunts: if platforms can (i) increase verification probability η and (ii) enforce escalating penalties that reduce continuation value after a false accusation, then even highly attention-seeking KOLs optimally refrain from low-evidence accusations, because doing so endangers future influence and revenue.

At the same time, the term $Bn_{A,1}$ highlights the tension: if engagement amplification is extremely strong, it can still dominate dynamic discipline. This implies a policy complementarity: escalating sanctions are most effective when combined with mechanisms that reduce mob amplification (lower B), so that the short-run gain from triggering a pile-on is not large enough to outweigh the long-run reputational cost.

10 Discussion and Policy Implications

Across the baseline and extensions, the core mechanism is stable: accusations and attacks can be sustained as equilibrium behavior when (i) the KOL's marginal payoff from engagement is high relative to expected penalties, and (ii) users face strong conformity and attention incentives, making attacking privately optimal when others are expected to attack.

The private-signal extension sharpens the interpretation of "consensus" in witch-hunts: a large observed mob does not necessarily reflect strong information about guilt; it may reflect the equilibrium selection induced by conformity incentives and amplification. Policy levers therefore correspond to moving parameters that enter the KOL and user thresholds: raising effective penalties (λD), reducing amplification (B), increasing friction for attacking (raising the effective cost of impulsive participation), and redesigning interfaces that magnify conformity payoffs.

11 Conclusion

This paper formalizes online accusations as a Bayesian game between an informed KOL and a continuum of users. The baseline model delivers clean best-response thresholds and clarifies how pooling and separating equilibria can sustain collective attacks under uncertainty. The main extension allows each user to observe a private signal, which endogenizes partial participation and highlights how information and coordination jointly shape the mass of attackers. Additional extensions with heterogeneous users, multiple KOLs, and dynamic reputational penalties preserve the same strategic logic while bringing the framework closer to online institutional realities. The resulting analysis provides a coherent foundation for understanding why digital witch-hunts persist and which structural levers can shift equilibrium behavior toward restraint.